

Municipal Infrastructure Damage Reporting App

Use-Case Flow Specification

UC-01 — Submit Damage Report

Primary Actor: Citizen Reporter

Supporting Actor: Municipal GIS System

Preconditions

1. The Citizen Reporter has access to the Municipal Infrastructure Damage Reporting App.
2. The Citizen Reporter has a photograph of the infrastructure damage.
3. The application is available for submitting a damage report.

Postconditions

1. The damage report is successfully recorded in the system.
2. The incident location is associated with the report.
3. The report is available for further processing by the relevant municipal department.

Main Success Scenario

1. The Citizen Reporter selects **Submit Damage Report**.
2. The system displays the damage-report form.
3. The Citizen Reporter selects the type of infrastructure damage.
4. The Citizen Reporter uploads a photograph of the damage.
5. The system extracts the EXIF GPS metadata from the photograph.
6. The system identifies the incident location using the extracted GPS coordinates.
7. The system displays the identified location on the municipal GIS map.
8. The Citizen Reporter confirms the report details.
9. The system records and submits the damage report.
10. The system routes the report to the relevant municipal engineering department.
11. The system displays a confirmation that the damage report has been submitted successfully.

Alternate Flow - A1. No Network Connectivity

1. The system detects that network connectivity is unavailable while the Citizen Reporter is submitting the report.
2. The system saves the drafted report locally.
3. The Citizen Reporter is informed that the report has been saved for later synchronization.
4. When network connectivity is restored, the system automatically synchronizes the saved report.
5. The system completes the report submission process.